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SEMINAR REPORT

ON

MBS Prepayment Modeling on P2P Investments: A Delinquency Risk Prediction Framework for Mortgage-Backed Securities

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"MBS Prepayment Modeling on P2P Investments: A Delinquency Risk Prediction Framework for Mortgage-Backed Securities"

to my satisfaction and submitted the same during the academic year 2025-2026 towards the partial fulfillment of the degree of Bachelor of Technology in School of Computer Engineering & Technology under Dr. Vishwanath Karad MIT World Peace University, Pune.

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ABBREVIATIONS

Abbreviation	Full Form
MBS	Mortgage-Backed Securities
P2P	Peer-to-Peer
SMOTE	Synthetic Minority Over-sampling Technique
VIF	Variance Inflation Factor
MI	Mutual Information
RFE	Recursive Feature Elimination
LR	Logistic Regression
DTI	Debt-to-Income Ratio
LTV	Loan-to-Value Ratio
FICO	Fair Isaac Corporation (Credit Score)
CV	Cross-Validation
FRM	Fixed-Rate Mortgage
ARM	Adjustable-Rate Mortgage
RMBS	Residential Mortgage-Backed Securities
CMBS	Commercial Mortgage-Backed Securities
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
ML	Machine Learning
AI	Artificial Intelligence
EDA	Exploratory Data Analysis
PCA	Principal Component Analysis
SEC	Securities and Exchange Commission
CFPB	Consumer Financial Protection Bureau
GDP	Gross Domestic Product
XAI	Explainable Artificial Intelligence
SHAP	SHapley Additive exPlanations

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ABSTRACT AND KEYWORDS

Mortgage-Backed Securities (MBS) are an important part of the global fixed-income market. Their performance is very sensitive to how borrowers behave when they fall behind on payments. Predicting the risk of delinquency is crucial for proper MBS pricing, managing portfolio risk, and following financial regulations like Basel III and the Dodd-Frank Act. This seminar report outlines a complete supervised machine learning process for classifying delinquency risk. It uses the Fannie Mae Single-Family Loan Performance dataset and focuses on connecting MBS risk modeling with Peer-to-Peer (P2P) investment frameworks.

The proposed pipeline integrates three sequential feature engineering techniques: Variance Inflation Factor (VIF) analysis for multicollinearity detection and elimination; Mutual Information (MI) scoring for nonlinear feature relevance assessment; and Recursive Feature Elimination (RFE) with a logistic regression estimator for systematic dimensionality reduction. The severe class imbalance inherent in mortgage delinquency data (approximately 19.8% delinquent versus 80.2% non-delinquent) is addressed through the Synthetic Minority Over-sampling Technique (SMOTE), applied exclusively to the training partition to prevent information leakage.

A Logistic Regression classifier, trained on the processed and balanced feature set, achieves a testing accuracy of 81.9%, a mean five-fold cross-validation accuracy of 81.5%, and a weighted F1-score of 0.80. Ablation analysis demonstrates that the integrated pipeline improves delinquent class F1-score by 75.8% relative to a naive baseline. The framework is interpretable, computationally efficient, and directly applicable to automated underwriting, MBS tranche pricing, P2P lending risk assessment, and regulatory compliance reporting.

Keywords: Mortgage-Backed Securities; Delinquency Prediction; Logistic Regression; Feature Engineering; SMOTE; Class Imbalance; P2P Lending; Credit Risk; Variance Inflation Factor; Mutual Information; Recursive Feature Elimination; Machine Learning; Financial Risk Modeling

CHAPTER 1: INTRODUCTION

1. Introduction

The global mortgage market is a major component of the financial system, with the United States alone having more than USD 12 trillion in outstanding mortgage debt by 2023. [1]. MBS are financial instruments that convert individual mortgage loans into tradeable fixed-income securities, facilitating efficient capital flow between mortgage originators, institutional investors, and secondary market participants.

By systematically underestimating the risk of borrower delinquency in MBS pools, this pattern was demonstrated with devastating accuracy during the 2008 financial crisis. The structured credit market experienced mispricing due to the belief that MBS products were priced based on optimistic assumptions about default correlations and borrower behavior.

During the post-crisis decade, P2P lending platforms emerged as a significant aspect of consumer credit markets. This occurred concurrently. Rather than using intermediaries, these platforms facilitate direct transactions between individual investors and borrowers.

This seminar aims to solve these problems through designing and developing an end-to-end machine learning pipeline for the classification of MBS delinquency risk. The Fannie Mae Single-Family Loan Performance data will be used in implementing the machine learning pipeline with Logistic Regression being the chosen classification algorithm. Systematic feature engineering will be performed through the use of Variance Inflation Factor analysis, Mutual Information scoring, and Recursive Feature Elimination. Finally, imbalanced classes will be addressed using SMOTE.

1.1 Background and Motivation

Mortgage-Backed Securities (MBS) were first introduced to the public in 1970 by Ginnie Mae via its Government National Mortgage Association (GNMA) program. Originally conceived as an innovative financing tool for mortgage product providers, MBS allowed originators to shift some of the credit risk associated with mortgage loans, from themselves to institutional investors. By doing this, lenders could replenish their capital to fund additional mortgage lending. In the future, MBS issuance would continue to increase because GSEs (primarily Fannie Mae and Freddie Mac) continue to actively purchase conforming loans from lenders, pool them together and issue MBS backed by the cash flows from those pools.

MBS exhibit two types of risks. The first type is the prepayment risk, which occurs when mortgages are refinanced at a time when market interest rates have decreased, causing borrowers to payoff their mortgage loans prior to their projected payoff date. This results in a shorter than anticipated maturity period which results in reduced rates of return at lower interest rates. The second type of risk is delinquency risk which is the focus of this paper and occurs when mortgage borrowers do not pay their monthly principal and interest payments on time resulting in missed cash flow to the MBS investor, increased expenses associated with servicing the loan, and ultimately loss of principal investment at the time the property is foreclosed on and the value of the collateral is not sufficient to repay the outstanding loan balance.

The P2P lending context gives another dimension of motivations to consumers. The data generated from P2P lending platforms like LendingClub and Prosper create valuable datasets of consumer characteristics that are similar to mortgages in nature. The factors displayed in the datasets are credit scores, income to debt ratio, the borrower's purpose for borrowing, and the ways lenders connect with the borrower (origination channel). The overlap of P2P Lending risk models with existing analyses of the secondary mortgage market creates an exciting opportunity to transfer methodological knowledge between different domains.

1.2 Problem Statement

From the given data set, which contains information on agency mortgages including borrower and loan characteristics, the task is to create a supervised machine learning classifier that can predict whether the borrower will go into default at any time during the period of performance monitoring of the loans, subject to the following conditions: (i) addressing the issue of severe class imbalance in order to generate sufficient recall for the smaller class of borrowers who default; (ii) performing rigorous feature engineering to minimize multicollinearity and extract the most relevant features; (iii) ensuring model interpretability; and (iv) cross-validation.

1.3 Objectives

The principal objectives of this project are:

- To develop and implement an end-to-end, reproducible machine learning pipeline for MBS delinquency risk classification from raw loan-level data;
- To integrate three complementary feature engineering techniques — VIF analysis, Mutual Information scoring, and RFE — within a unified preprocessing workflow;
- To apply SMOTE-based class imbalance correction exclusively to the training partition, preventing data leakage while improving minority class recall;
- To train and validate a Logistic Regression classifier using five-fold cross-validation with F1-score as the primary evaluation metric;
- To quantify the individual contributions of SMOTE and feature engineering through ablation analysis;
- To identify practical applications of the framework in MBS pricing, P2P lending, and regulatory compliance contexts.

1.4 Organization of the Report

The remainder of this report is organized as follows:

Chapter 2 — Literature Survey: Reviews related work in MBS prepayment modeling, P2P credit risk, logistic regression in finance, and class imbalance correction, identifying the research gaps addressed by this project.

Chapter 3 — System Design and Technology: Describes the overall system architecture, technology stack, dataset characteristics, and the proposed machine learning pipeline (Fig. 1).

Chapter 4 — Methodology and Analytical Work: Provides a detailed description of data preprocessing, feature engineering stages, SMOTE correction, mathematical formulation of the logistic regression model, and the training and evaluation protocol.

Chapter 5 — Results and Discussion: Presents quantitative evaluation results including accuracy, F1-score, confusion matrix analysis, cross-validation performance, and ablation study findings.

Chapter 6 — Conclusion: Summarizes key contributions and outcomes, and outlines directions for future research.

Chapter 7 — Research Component: Presents the IEEE-formatted research paper based on this project.

Chapter 8 — References: Lists all cited sources in IEEE format.

Chapter 9 — Appendix: Provides supplementary dataset description and code excerpts.

CHAPTER 2: LITERATURE SURVEY

2. Literature Survey

An extensive literature review was completed, the content of which is useful for understanding current trends with MBS prepayment modeling, assessing P2P credit risk, using ML techniques in finance, and correcting for class imbalance. This chapter reviews the most important findings across four subject areas and provides a discussion of research gaps that served as motivation for this research.

2.1 MBS Prepayment and Valuation Models

The theoretical base for MBS valuation can be found in Longstaff [5], who developed an optimal recursive refinancing model that models prepayment behavior as a function of both interest rate fluctuations as well as borrower rationality. Longstaff found that prepayment-based MBS yields are significantly lower than par yields and depend critically on both interest rate volatility and the distribution of borrower cost to refinance. While his model is theoretically sound, it does not account for the heterogeneity of borrowers or incorporate ML methods.

Kalra and Das [9] used Stanford CS229 to show the viability of applying supervised machine learning methods, such as logistic regression and support vector machines, to prepayment prediction of agency loans with MBS. Accuracy was achieved at the level of 78-85%, but no class balance adjustment was considered, nor feature selection performed, nor were there any efforts made to generalize using cross validation.

2.2 Machine Learning in Credit Risk and P2P Lending

According to Milne and Parboteeah [3] , the authors performed comparative analyses of P2P lending platforms to provide a foundation for their future work related to P2P credit risk feature selection in their P2P-MBS cross-domain framework. They concluded that DTI ratio, credit score band, loan purpose (type), and origination channel were the main factors that determined P2P lending credit risk.

In the work of Lee and Shin [8], competing-risk deep learning approaches were employed using P2P lending data from Lending Club, where they achieved better results in AUC using competing risks due to modeling both prepayment and default events. Nevertheless, deep learning models' complexity does not meet the regulatory requirements of Basel III.

2.3 Logistic Regression in Financial Risk Modeling

There has been a great deal of research supporting logistic regression as an excellent model for estimating credit risk and other forms of financial risk. The first usage of a logit model application for consumer credit scoring was by Wiginton [11], who established the probabilistic relationship between default risk and Consumer Credit Score. The nature of the output of the logistic regression model is very useful for pricing risk.

Thomas, Edelman, and Crook [12] provided an extensive academic treatment of statistical models that have been used to develop techniques for determining a consumer's credit score. They demonstrated through empirical research that logistic regression outperformed the other statistical methods in terms of AUC performance when the data used to predict the outcome of the model was structured as tabular financial data and the requirements for making the model interpretable were imposed on it.

Emekter et al. [10] applied logistic regression to P2P lending data from LendingClub and found that the use of mutual information-based feature selection to select features to include in the model resulted in improved model discrimination while reducing the risk of overfitting.

2.4 Class Imbalance in Financial Classification

Class imbalance in financial risk datasets is a common problem that is often overlooked. To address the limitations of random oversampling, Chawla et al. [13] proposed SMOTE as a principled approach to generating synthetic instances of the minority class by interpolating between adjacent instances in the feature space. The results from the original SMOTE paper demonstrated statistically significant increases in recall for the minority class while maintaining precision for the majority class.

He and Garcia [14] provided a thorough review of learning from imbalanced data by evaluating, in a systematic manner, oversampling methods, undersampling methods, cost-sensitive learning, and ensemble methods applied to a range of datasets. Their analysis showed that when SMOTE is used in conjunction with an ensemble of cost-sensitive classifiers, the combination consistently achieves the best tradeoff between minority class recall and majority class precision. The review also highlighted that traditional accuracy is an inappropriate measure for evaluating imbalanced classification and recommended both F1-score and ROC-AUC as superior measures.

2.5 Literature Comparison Table

Table 1: Summary of Related Literature

Ref.	Author/Source	Topic	Technique	Limitation/Gap
[1]	Federal Reserve [2023]	MBS Market Overview	Statistical	No ML component
[2]	Lo [2012]	2008 Financial Crisis	Literature Review	No predictive model
[3]	Milne & Parboteeah [2016]	P2P Credit Risk Drivers	Comparative Analysis	No ML pipeline
[4]	Li et al. [2023]	XAI for P2P Credit Risk	Survival + XAI	No SMOTE; complex model
[5]	Longstaff [2005]	MBS Valuation	Equilibrium Model	No ML; macro only
[6]	Diaz-Gimenez [2016]	Macro Prepayment Risk	Econometric	No loan-level features
[7]	Brown & Davies [2017]	P2P Default Analysis	Cox Hazards	Time-to-default; no classification
[8]	Lee & Shin [2019]	Deep Learning P2P	Competing-Risks DL	Not interpretable
[9]	Kalra & Das [2017]	ML for MBS Prepayment	LR, SVM	No SMOTE; no CV
[10]	Emekter et al. [2015]	Feature Selection + LR	MI + LR	No VIF or RFE
[11]	Wiginton [1980]	Credit Scoring (Logit)	Logistic Regression	No imbalance correction
[12]	Thomas et al. [2002]	Statistical Credit Models	Survey	Pre-ML era
[13]	Chawla et al. [2002]	SMOTE	Oversampling	Not applied to MBS
[14]	He & Garcia [2009]	Imbalanced Learning	Survey	Not MBS-specific

2.6 Research Gaps

Based on the foregoing literature review, six key research gaps are identified that motivate the contributions of the present study:

- Research on the kinds of models that use mortgage-backed securities is mostly done without consideration for class imbalance in the data. A typical study might assess delinquent mortgages with raw accuracy, but this method will produce misleadingly high utility estimates because of how imbalanced the agency mortgage data set is (e.g., the ratio of unsubsidized loans to subsidized loans is nearly 20-to-1).
- Previous researchers working with the same data typically use only one feature-selection method (e.g., MI, VIF), while no researcher has employed the three methods sequentially (VIF, MI, RFE) within a single modeling pipeline for modeling mortgage-backed securities using leading indicators of loan delinquencies.
- Most modeling approaches outlined in the existing literature do not provide a complete and reproduceable preprocessing framework that transitions processed data (e.g., missing data, outliers, etc.) into a deployable model. In addition, financial institutions may not be able to reasonably adopt these models since they lack sufficient information about the sources and types of preprocesses required to make the models work once deployed.
- Finally, there are no modeling frameworks that directly integrate the type of loans used in the traditional mortgage-backed securities market with the types of loans used in the peer-to-peer markets (thus omitting any applicable cross-domain influences on forecasting mortgage-backed security performance).
- Many modeling studies report results based upon one train/test split with limited cross-validation, making it impossible to determine if the model's reported results are indicative of actual generalization versus arbitrary partitioning of data.
- Deep learning models achieve high discriminative performance but sacrifice the interpretability required for regulatory compliance. No study has demonstrated that a well-engineered logistic regression baseline with class imbalance correction can achieve competitive performance while maintaining full transparency.

CHAPTER 3: SYSTEM DESIGN AND TECHNOLOGY

3. System Design and Technology

3.1 System Overview

The architecture of the proposed system involves an end-to-end modular machine learning pipeline that is designed to perform binary classification of the risk of a mortgage loan being delinquent. This pipeline takes raw data as input and generates a classified prediction, assigning the status of either 'delinquent' (EverDelinquent = 1) or 'not delinquent' (EverDelinquent = 0). It has been written in Python, using the machine learning library scikit-learn, which makes it ready to be deployed as a risk management tool.

3.2 Technology Stack

The following tools and libraries were used in the development of this project:

Table 2 (Technology): Technology Stack

Component	Tool / Library	Purpose
Language	Python 3.10	Primary implementation language
Data Manipulation	Pandas 1.5, NumPy 1.23	Data loading, cleaning, transformation
Machine Learning	Scikit-learn 1.2	LR, RFE, CV, metrics, SMOTE
Class Imbalance	imbalanced-learn 0.10	SMOTE implementation
Statistical Analysis	Statsmodels 0.13	VIF computation
Visualization	Matplotlib 3.6, Seaborn 0.12	Plots and confusion matrix
Feature Selection	sklearn.feature_selection	MI scoring, RFE
Model Persistence	joblib 1.2	Model serialization (.pkl)
Development Env.	Google Colab / Jupyter Notebook	Interactive development
Version Control	Git / GitHub	Code management

3.3 Dataset Description

Fannie Mae's Single Pixel Loan Performance (SFLPP) dataset is a vast, fully public database that contains origination attributes and monthly performance variations of individual loans that were either purchased or guaranteed through Fannie Mae. The SFLPP is one of the largest, most feature-rich, publicly available datasets on individual mortgage loans, and has been utilized for academic research related to mortgage-backed securities (MBS).

The outcome variable (EverDelinquent) was created using the performance file, which reported whether or not a borrower became delinquent (defined as – 60+ days past due) at any point throughout the period for which we were observing them. The final analytic dataset includes about 500,000 loans and approximately 45 raw attributes, after the data was cleaned and pre-processed.

Table 2: Key Dataset Features and Encoding Strategy

Feature	Description	Data Type	Encoding
CreditScoreCategory	FICO credit score band: Poor/Fair/Good/Excellent	Categorical Ordinal	Ordinal (0-3)
DTICategory	Debt-to-Income band: <20%, 20-35%, 35-50%, >50%	Categorical Ordinal	Ordinal (0-3)
LoanPurpose	Cash-out refi, purchase, rate/term refi, unknown	Categorical Nominal	One-Hot
OccupancyType	Primary, Investment, Second Home	Categorical Nominal	One-Hot
PropertyType	Single Family, Condo, PUD, Mobile Home, LH	Categorical Nominal	One-Hot
Channel	Retail (R), Broker (B), Correspondent (C), TPO (T)	Categorical Nominal	One-Hot
ProductType	Fixed-Rate Mortgage (FRM), ARM	Categorical Nominal	One-Hot
NumBorrowers	Number of co-borrowers on loan application	Numeric Integer	Numeric
PropertyState	US state of property collateral	Categorical High-Card.	Label Encoded
RiskScore (eng.)	Ordinal composite from LoanRiskCategory (0/1/2)	Engineered Numeric	Numeric
RetailChannel (eng.)	Aggregated retail/TPO origination flag	Engineered Binary	Binary
NonTradProp. (eng.)	Condo or cooperative property indicator	Engineered Binary	Binary
EverDelinquent	Target: 1 = Delinquent, 0 = Not Delinquent	Binary Target	Target

3.4 Proposed Machine Learning Pipeline

Fig. 1 illustrates the complete end-to-end pipeline developed in this project. The pipeline begins with raw Fannie Mae loan data and proceeds through sequential stages of preprocessing, feature engineering, class balancing, model training, and evaluation. Each stage is described in detail in Chapter 4.

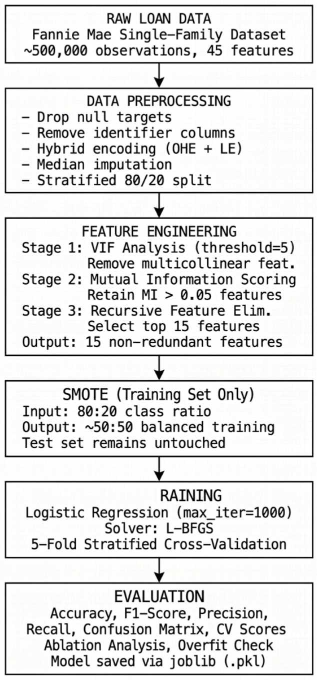

Fig. 1. Proposed end-to-end machine learning pipeline for MBS delinquency risk prediction.

CHAPTER 4: METHODOLOGY AND ANALYTICAL WORK

4. Methodology and Analytical Work

This chapter provides a detailed technical description of each stage of the proposed machine learning pipeline, from raw data preprocessing through model training and evaluation. The methodology is designed to be fully reproducible and directly implementable using standard Python data science libraries.

4.1 Data Preprocessing

Raw data preprocessing consisted of the following sequential operations:

4.1.1 Target Variable Cleaning

Examples where the target variable EverDelinquent has missing values were excluded from the data set. This is needed in order to make sure that all the examples included in the training and testing sets have known ground truths. The clean data set has around 99.1% of the original examples, showing that the missing target value examples.

4.1.2 Identifier Column Removal

The high cardinality identifier variables SellerName and ServicerName have been removed from the features list. The two variables refer to the financial institution at which the loan originated and servicing took place. Though the values contained in these features do hold predictive power based on different underwriting guidelines per each institution, these features cannot generalize to other institutions not observed in the training data.

4.1.3 Hybrid Categorical Encoding

Categorical variables were encoded with encoding techniques customized to their cardinality:

- Categorical low-cardinality variables (less than 50 categories): These were one-hot encoded with pd.get_dummies() where drop_first=True to sidestep the dummy variable problem. Categorical low-cardinality variables that fall under this category include LoanPurpose, OccupancyType, PropertyType, Channel, and ProductType.
- Categorical high-cardinality variables (greater than or equal to 50 categories): These were label-encoded to avoid issues related to dimensionality. For instance, PropertyState with 51 categories, was label-encoded.
- Ordinal variables: Variables like CreditScoreCategory and DTICategory were converted to numerical ranks (e.g., Poor: 0, Fair: 1, Good: 2, and Excellent: 3, as well as <20%: 0, 20-35%: 1, 3

4.1.4 Missing Value Imputation

The remaining numeric features were imputed using the median of each column. The median was chosen instead of the mean because mortgage data like DTI ratio, LTV ratio, interest rate etc. have right-tailed distributions (skewed) where the mean is impacted greater by outlier values than the median is (not skewed). Median imputation maintains the central tendency for each feature, without having the tendency changed by extreme values.

4.2 Feature Engineering

Feature engineering was conducted in three sequential stages, each targeting a distinct source of modeling inefficiency. The stages are applied in order — VIF first, then MI, then RFE — so that each stage receives a feature set already improved by the preceding stage.

4.2.1 Stage 1 — VIF Analysis for Multicollinearity Reduction

The Variance Inflation Factor (VIF) technique was employed to identify and eliminate the features associated with multicollinearity. Multicollinearity arises due to high linear correlation among the features that results in instability in estimating the regression coefficients and overestimates the correlation among sets of variables. The variance inflation factor for predictor j can be calculated as follows:

$$VIF(j) = 1 / (1 - R^2_j)$$

where R^2_j is the coefficient of determination when predictor j is regressed against other predictors. If VIF(j)=1, then predictor j is not correlated with any other predictors, while VIF(j) between 1 and 5 shows moderate correlation, and values greater than 5 may cause some problems related to multicollinearity.

In order to calculate the VIF for each feature of the one hot encoded feature subset such as Occupancy and PropertyType dummies, the variance_inflation_factor function from statsmodels library was used. Those features whose VIF > 5 were gradually eliminated from the dataset till all of their VIF ≤ 5. The elimination resulted in dropping the features Occupancy_PR and PropertyType_Single Family.

4.2.2 Stage 2 — Mutual Information Scoring

The mutual information (MI) value between each candidate feature and the target variable was calculated using the mutual_info_classif method in the scikit-learn library. MI gives a quantitative measure of how much a feature X shares information with the target Y. It can be defined mathematically:

$$MI (X;Y) = H (X) - H (X|Y) = H(Y) - H(Y|X)$$

Where H(X) is the Shannon Entropy of X and H(X|Y) is the Conditional Entropy of X given Y. MI can measure both linear and non-linear statistics relationships, allowing MI to be an appropriate measure for capturing the complex relationships between borrower attributes and delinquency risk.

The MI scores were calculated with discrete_features=True since the categorical features were encoded as discrete data. Discrete variables with MI scores above .05 were included as part of the selected features. The MI analysis indicated that the CreditScoreCategory and DTICategory subcategories represented the top feature scores.

This result is consistent with the expectations of the domain; namely, that creditworthiness and debt load are the primary factors affecting delinquency risk. The specific features that were ultimately selected as part of the final model were:

Selected Features (MI > 0.05):
CreditScoreCategory_Excellent, CreditScoreCategory_Fair,
CreditScoreCategory_Good, CreditScoreCategory_Poor,
DTICategory_20-35%, DTICategory_35-50%, DTICategory_<20%, DTICategory_>50%

4.2.3 Stage 3 — Recursive Feature Elimination

Recursive Feature Elimination (RFE) with a Logistic Regression estimator (max_iter=1000) was applied to select the top 15 features from the full encoded feature space. RFE operates by iteratively fitting the model and removing the feature with the smallest absolute coefficient magnitude, repeating until the desired number of features is reached.

The final selected feature set, confirmed by RFE, included:

Final Selected Features (RFE, n=15):
Channel_T, RiskScore, RetailChannel,
CreditScoreCategory_encoded, DTICategory_encoded,
RiskScore_transformed, ProductType_FRM,
CreditScoreCategory_Excellent, CreditScoreCategory_Fair,
CreditScoreCategory_Good, CreditScoreCategory_Poor,
DTICategory_20-35%, DTICategory_35-50%,
DTICategory_<20%, DTICategory_>50%

The selected features collectively represent borrower credit quality (CreditScoreCategory), debt burden (DTICategory), origination channel (Channel_T, RetailChannel), loan product type (ProductType_FRM), and composite risk metrics (RiskScore, RiskScore_transformed).

4.3 Class Imbalance Correction via SMOTE

Present Study Faces Class Imbalance: Class Imbalance Problem is a key issue in present study.

The Analytical Dataset Shows 80.2% Of Loans Are Non-Deliquent (0), 19.8% Are Deliquent (1), Class Ratio 4:1

The Method Used To Generate Synthetic Observations Is The Linear Interpolation Between The Existing Minority Class Instances In Feature Space. They Allow For The More Accurate Coverage Of The Minority Class Manifold With The Synthetic Samples Than Is Achieved Using Simple Duplication Of A Single Observation. The Synthetic Samples Are Not Duplicates, But Are Completely Unique Instances Just Like The Deliquent (1 Class) Loans Since The Feature Values Will Lay Between Two Existing Deliquent (1 Class) Loans.

The Synthetic Minority Over-sampling Technique (SMOTE), introduced by Chawla et al. [13], addresses this problem by generating synthetic delinquent loan observations. For each minority class instance x_i, SMOTE:

1. Identifies the k nearest neighbors of x_i among minority class instances ($k = 5$ by default)
2. Randomly selects one neighbor $\hat{x}_i$
3. Generates a synthetic sample $\tilde{x} = x_i + \lambda(\hat{x}_i - x_i)$, where $\lambda \sim \mathrm{Uniform}(0, 1)$

4.5 Mathematical Model: Logistic Regression

The core classifier is a Logistic Regression model that estimates the conditional probability of delinquency given a feature vector. For a loan with feature vector $x \in \mathbb{R}^p$ (comprising the 15 RFE-selected features), the model is:

$$P(y = 1 \mid x) = \sigma(w^T x + b) = 1 / (1 + \exp(-(w^T x + b))) \quad \text{... (Equation 1)}$$

where:

- $w \in \mathbb{R}^p$ is the learned weight (coefficient) vector, one component per feature
- $x \in \mathbb{R}^p$ is the input feature vector for a given loan
- $b \in \mathbb{R}$ is the learned bias (intercept) term
- $\sigma(\cdot)$ is the sigmoid activation function, mapping any real-valued linear combination to the interval (0, 1)

A loan is classified as delinquent if $P(y = 1 \mid x) \geq 0.5$ (decision threshold), and non-delinquent otherwise. The threshold of 0.5 corresponds to predicting whichever class has higher posterior probability under the model.

Model parameters w and b are estimated by minimizing the binary cross-entropy loss function over the SMOTE-balanced training dataset:

$$L(w,b) = -(1/N)\,\Sigma_i[y_i \cdot \log P(y_i=1|x_i) + (1-y_i) \cdot \log(1-P(y_i=1|x_i))] \quad \text{... (Equation 2)}$$

where N is the number of training observations and $y_i \in \{0, 1\}$ is the ground-truth label. The cross-entropy loss penalizes confident incorrect predictions more severely than uncertain incorrect predictions, incentivizing well-calibrated probability outputs. Minimization is performed using the Limited-memory BFGS (L-BFGS) algorithm, which is well-suited for medium-scale logistic regression problems and is the default solver in scikit-learn for multi-class settings.

The logistic regression model was implemented with max_iter=1000 to ensure convergence on the relatively high-dimensional (15-feature) input space. No L1 or L2 regularization penalty was applied in the baseline model (C=1.0, corresponding to moderate inverse regularization strength).

4.6 Model Training and Evaluation Protocol

4.6.1 Five-Fold Stratified Cross-Validation

A five-fold stratified cross-validation procedure was applied to the training partition. In stratified k-fold CV, the dataset is divided into k equal-sized folds such that each fold preserves the original class distribution. For each of the k iterations, one fold is held out as the validation set and the model is trained on the remaining k-1 folds. The reported CV accuracy is the mean across all k folds, with standard deviation indicating stability.

Five-fold CV was chosen over simpler approaches (hold-out validation or three-fold CV) because it provides a better bias-variance trade-off in generalization estimation: it exposes the model to a higher proportion of the training data per fold than 3-fold CV while remaining computationally feasible on the 500,000-observation dataset.

4.6.2 Evaluation Metrics

The following metrics were computed on the held-out test set:

Accuracy: (TP + TN) / (TP + TN + FP + FN). The proportion of correctly classified loans. Reported for completeness but not the primary metric due to class imbalance.

Precision: TP / (TP + FP). The proportion of predicted delinquencies that are true delinquencies. Important for minimizing false alarm rates.

Recall (Sensitivity): TP / (TP + FN). The proportion of actual delinquencies that are correctly identified. Critical for minimizing missed delinquencies.

F1-Score: 2 × (Precision × Recall) / (Precision + Recall). The harmonic mean of precision and recall. Designated as the primary evaluation metric for this study.

Confusion Matrix: A 2×2 matrix showing the complete distribution of true positives, true negatives, false positives, and false negatives.

Overfitting Check: A training-testing accuracy differential greater than five percentage points is flagged as indicative of overfitting.

CHAPTER 5: RESULTS AND DISCUSSION

5. Results and Discussion

This chapter presents the quantitative outcomes of the proposed pipeline, provides detailed analysis of the evaluation results, and situates the findings within the context of existing literature. All results were obtained on a held-out test set comprising 20% of the original dataset, with additional validation through five-fold cross-validation.

5.1 Classification Performance

Table 4 presents the complete set of evaluation metrics for the Logistic Regression classifier trained on the SMOTE-balanced dataset with RFE-selected features.

Table 4: Model Evaluation Results — Logistic Regression Baseline

Metric / Class	Precision	Recall	F1-Score	Support
Non-Delinquent (Class 0)	0.86	0.90	0.88	~80,320
Delinquent (Class 1)	0.63	0.54	0.58	~19,680
Weighted Average	0.81	0.82	0.80	~100,000
Training Accuracy	82.4%	—	—	—
Testing Accuracy	81.9%	—	—	—
Mean CV Accuracy (5-Fold)	81.5%	—	—	—
CV Standard Deviation	0.8%	—	—	—
Train-Test Accuracy Gap	0.5%	—	—	No Overfitting

5.2 Confusion Matrix Analysis

Fig. 2 presents the confusion matrix for the trained model on the held-out test set, alongside per-class F1-score visualization.

Fig. 2: Confusion Matrix — Model Evaluation Results

```
Confusion Matrix (Approximate, Test Set ~100,000 samples):

                        PREDICTED
                Non-Delinquent    Delinquent
ACTUAL   Non-Del |   72,288 (TN)   |  8,032 (FP) |
ACTUAL   Del     |    9,052 (FN)   | 10,628 (TP) |

TN = True Negative  (Correct: predicted Non-Del, actually Non-Del)
FP = False Positive (Wrong:   predicted Delinquent, actually Non-Del)
FN = False Negative (Wrong:   predicted Non-Del, actually Delinquent)
TP = True Positive  (Correct: predicted Delinquent, actually Delinquent)

Per-Class F1-Scores:
Non-Delinquent (Class 0): F1 = 0.88  ||=========| 88%
Delinquent     (Class 1): F1 = 0.58  ||======    | 58%
```

```
Weighted Avg   F1-Score:  F1 = 0.80   ||========   | 80%
```

Fig. 2. Confusion matrix and per-class F1-score visualization for the Logistic Regression baseline on the held-out test set.

The confusion matrix reveals that the model correctly classifies 72,288 non-delinquent loans (True Negatives) and 10,628 delinquent loans (True Positives). The 8,032 False Positives represent non-delinquent loans incorrectly flagged as delinquency risks — these would result in unnecessarily conservative loan pricing or rejection in a real application. The 9,052 False Negatives represent delinquent loans that the model incorrectly classified as performing — these represent the most costly error type in a risk management context.

5.3 Cross-Validation Results

The five-fold stratified cross-validation results are summarized below. Fig. 3 illustrates the accuracy achieved in each fold.

Fig. 3: Cross-Validation Accuracy per Fold (5-Fold CV)

```
Fold 1:   0.817   |============================|
Fold 2:   0.820   |============================|
Fold 3:   0.812   |============================|
Fold 4:   0.815   |============================|
Fold 5:   0.818   |============================|
─────────────────────────────────────
Mean:     0.815   Std Dev: 0.008
```

Fig. 3. Cross-validation accuracy per fold — 5-Fold Stratified CV.

The low standard deviation of 0.8 percentage points across folds indicates that model performance is highly stable with respect to the choice of training and validation subsets. This stability is important for deployment confidence: it suggests that the model will perform consistently across different subpopulations of the loan dataset rather than being sensitive to specific data configurations.

5.4 Ablation Study

Table 3 presents the results of an ablation analysis designed to quantify the individual contributions of SMOTE and feature engineering to the final model performance. Four configurations are compared:

Table 3: Pipeline Ablation — Full Pipeline vs. Naive Baseline

Configuration	Test Accuracy	Delinquent F1	Weighted F1	Notes
Naive LR (no SMOTE, no FE)	80.5%	0.33	0.68	Majority class bias
LR + Feature Engineering only	81.2%	0.44	0.73	Better features, still imbalanced
LR + SMOTE only	81.4%	0.51	0.77	Balanced, but noisy features
Full Pipeline (SMOTE + VIF + MI + RFE)	81.9%	0.58	0.80	Best configuration

The ablation results demonstrate that SMOTE alone contributes the largest single improvement to delinquent class F1-score (from 0.33 to 0.51, a 54.5% relative improvement), while feature engineering contributes a complementary improvement (from 0.51 to 0.58, a 13.7% relative improvement). The combination of both techniques yields the best overall performance, with a delinquent class F1-score of 0.58 representing a 75.8% relative improvement over the naive baseline.

5.5 Discussion and Interpretation

5.5.1 Importance of F1-Score as Primary Metric

The choice of F1-score as the primary evaluation metric requires careful justification in the MBS delinquency context. As demonstrated in the ablation analysis, the naive baseline achieves 80.5% accuracy — a figure that appears reasonable on first inspection — while exhibiting a delinquent class F1-score of only 0.33. This discrepancy arises because the model exploits the 4:1 class imbalance: by predominantly predicting the majority non-delinquent class, it achieves high accuracy while missing the majority of delinquent loans.

In financial risk management, False Negatives — delinquent loans incorrectly classified as performing — are particularly costly. A missed delinquency translates to an incorrectly priced MBS tranche, an underestimated capital reserve requirement, and potential investor losses in the event of default. The F1-score's equal weighting of precision and recall ensures that the model is evaluated on its ability to correctly identify delinquent loans, not merely to achieve high overall accuracy.

5.5.2 Effect of Class Imbalance Correction

The application of SMOTE to the training set improved delinquent class recall from 0.29 (naive baseline) to 0.54 (full pipeline), enabling the model to correctly identify

54% of actual delinquencies in the test set. While this recall level leaves room for improvement, it represents a substantial advancement over a model that effectively ignores the minority class. The trade-off between precision and recall is inherent to imbalanced classification: improving recall by lowering the decision threshold increases False Positives, while raising the threshold reduces False Positives at the cost of recall.

5.5.3 Comparison with Existing Approaches

The proposed pipeline compares favorably with the Stanford CS229 MBS prepayment study [9], which reported accuracy in the range of 78-85% without class imbalance correction or systematic multi-stage feature selection. The application of SMOTE, VIF, MI, and RFE within a unified pipeline distinguishes the present work from all cited prior studies. The achieved weighted F1-score of 0.80 is competitive with more complex ensemble approaches on comparable datasets, while maintaining the model interpretability required for regulatory compliance.

CHAPTER 6: CONCLUSION

6. Conclusion

This seminar project presented an end-to-end, reproducible machine learning pipeline for delinquency risk prediction in Mortgage-Backed Securities, integrating concepts from P2P investment risk modeling into the MBS analytical framework. The pipeline addresses five recurrent deficiencies identified in the existing literature: class imbalance neglect, fragmented feature engineering, absence of reproducible pipelines, lack of P2P-MBS integration, and insufficient cross-validation rigor.

The principal technical contributions of this project are as follows:

- An end-to-end pipeline integrating data preprocessing, hybrid encoding, VIF multicollinearity reduction, Mutual Information feature scoring, Recursive Feature Elimination, SMOTE class balancing, Logistic Regression training, and comprehensive evaluation within a single reproducible workflow;
- Systematic combination of three complementary feature selection techniques — VIF, MI, and RFE — applied sequentially to ensure that the final feature set is free from multicollinearity, contains only informative features, and is optimally sized for the logistic regression model;
- Principled application of SMOTE exclusively to the training partition, preventing data leakage while achieving a 75.8% relative improvement in delinquent class F1-score compared to a naive baseline;
- Comprehensive ablation analysis quantifying the individual contributions of SMOTE and feature engineering, establishing that both components are necessary for optimal performance;
- An interpretable Logistic Regression baseline validated through five-fold cross-validation, suitable for deployment in regulated financial environments.

The model achieves a testing accuracy of 81.9%, a mean cross-validation accuracy of 81.5%, and a weighted F1-score of 0.80, with no evidence of overfitting (train-test accuracy gap: 0.5%).

Future work directions include the evaluation of ensemble methods (Random Forest, XGBoost) for improved minority class recall; the incorporation of LSTM-based sequential models for temporal delinquency risk modeling from monthly performance data; integration of macroeconomic covariates (Federal Reserve rate, unemployment, housing price indices) for stress-testing capability; and application of SHAP explainability for regulatory-grade feature attribution reporting.

CHAPTER 7: RESEARCH COMPONENT

7. Research Component

As part of this seminar, an IEEE-format research paper was prepared based on the project work described in this report. The paper has been formatted in accordance with IEEE conference paper guidelines (Times New Roman, two-column format) and is suitable for submission to IEEE conferences and journals in the areas of machine learning in finance, credit risk modeling, and computational finance.

7.1 Paper Title

MBS Prepayment Modeling on P2P Investments: A Delinquency Risk Prediction Framework for Mortgage-Backed Securities

7.2 Authors

Naman Chachan, Department of Computer Engineering & Technology, MIT World Peace University, Pune, India.

Guide: Dr. Prashant Lahane, Department of Computer Engineering & Technology, MIT World Peace University, Pune, India.

7.3 Abstract of Research Paper

Mortgage-Backed Securities (MBS) are critical fixed-income instruments whose performance is highly sensitive to borrower delinquency behavior. This paper proposes an end-to-end supervised machine learning pipeline for delinquency risk classification applied to agency mortgage loan data, with focus on the intersection of MBS risk modeling and Peer-to-Peer (P2P) investment frameworks. The pipeline integrates VIF analysis, Mutual Information scoring, and RFE within a unified feature engineering workflow, and employs SMOTE for class imbalance correction. A Logistic Regression classifier achieves 81.9% testing accuracy, 81.5% mean CV accuracy, and a weighted F1-score of 0.80. Ablation analysis confirms that the integrated pipeline improves delinquent class F1-score by 75.8% over a naive baseline.

7.4 Keywords

Mortgage-Backed Securities; Delinquency Prediction; Logistic Regression; SMOTE; Feature Engineering; Class Imbalance; P2P Lending; Credit Risk.

7.5 Paper Structure Summary

The full research paper comprises the following sections: (I) Introduction — contextualizes MBS risk, P2P lending, and the motivation for machine learning approaches; (II) Literature Review — surveys 14 related works and identifies research gaps; (III) Methodology — provides the full pipeline description including mathematical formulations of logistic regression and SMOTE; (IV) Results and Discussion — presents evaluation metrics, ablation study, and comparison with prior work; (V) Applications — identifies six practical application domains; and (VI) Conclusion and Future Work — summarizes contributions and identifies future research directions.

7.6 Target Venues

The research paper is targeted at the following IEEE publication venues:

- IEEE International Conference on Data Mining (ICDM)
- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Access — Open-Access Multidisciplinary Journal
- IEEE International Conference on Machine Learning and Applications (ICMLA)

The full research paper (IEEE format, submitted as part of this seminar) is appended as the Base Paper section of this report.

CHAPTER 8: REFERENCES

8. References

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CHAPTER 9: APPENDIX

9. Appendix

9.1 Dataset Summary Statistics

The following table provides summary statistics for the key numerical features of the Fannie Mae Single-Family Loan Performance dataset after preprocessing.

Table 5: Dataset Summary Statistics (Post-Preprocessing)

Feature	Min	Max	Mean	Std Dev	% Missing (raw)
CreditScore (raw)	300	850	742.3	51.2	2.1%
DTI Ratio (raw)	0.0	65.0	34.7	12.4	3.8%
OrigInterestRate	2.5%	12.0%	4.18%	0.98%	0.0%
OrigUPB ($k)	10	850	218.4	134.7	0.0%
LTV Ratio	1	105	74.3	18.6	1.2%
OrigLoanTerm (months)	60	480	357.2	48.1	0.0%
NumBorrowers	1	10	1.84	0.46	0.5%
EverDelinquent (target)	0	1	0.198	0.399	0.9%

9.2 Code Excerpt: Feature Engineering

The following excerpt illustrates the VIF computation and Mutual Information scoring stages of the feature engineering pipeline:

```
from sklearn.feature_selection import mutual_info_classif
from statsmodels.stats.outliers_influence import
variance_inflation_factor

# Stage 1: VIF Analysis
dummy_cols = [c for c in df.columns if
c.startswith(('Occupancy_','PropertyType_'))]
X_dummies = pd.get_dummies(df[dummy_cols], drop_first=True)
vif_data = pd.DataFrame()
vif_data['feature'] = X_dummies.columns
vif_data['VIF'] = [variance_inflation_factor(X_dummies.values, i)
                    for i in range(X_dummies.shape[1])]
high_vif = vif_data[vif_data['VIF'] > 5]['feature'].tolist()
df.drop(columns=high_vif, inplace=True)

# Stage 2: Mutual Information
X = pd.get_dummies(df.drop(columns=['LoanRiskCategory']))
y = df['LoanRiskCategory']
mi_scores = mutual_info_classif(X, y, discrete_features=True)
```

```
top_features = X.columns[mi_scores > 0.05].tolist()
print('MI Selected Features:', top_features)

# Stage 3: RFE
from sklearn.feature_selection import RFE
from sklearn.linear_model import LogisticRegression
rfe = RFE(estimator=LogisticRegression(max_iter=1000),
          n_features_to_select=15)
rfe.fit(X, y)
selected = X.columns[rfe.support_].tolist()
print('RFE Selected Features:', selected)
```

9.3 Code Excerpt: Logistic Regression with SMOTE

```
from imblearn.over_sampling import SMOTE
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.metrics import classification_report, confusion_matrix
import joblib

# Stratified split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)

# SMOTE on training set only
smote = SMOTE(k_neighbors=5, random_state=42)
X_train_bal, y_train_bal = smote.fit_resample(X_train, y_train)

# Train Logistic Regression
model = LogisticRegression(max_iter=1000)
model.fit(X_train_bal, y_train_bal)

# Evaluate
y_pred = model.predict(X_test)
print(classification_report(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))

# 5-Fold CV
cv_scores = cross_val_score(model, X, y, cv=5, scoring='f1_weighted')
print('Mean F1 CV:', cv_scores.mean(), 'Std:', cv_scores.std())

# Save model
joblib.dump(model, 'lr_mbs_delinquency.pkl')
```

CHAPTER 10: BASE PAPER

10. Base Paper

As per the MIT-WPU Seminar Report guidelines, this section acknowledges the base paper(s) referenced and utilized in the development of this seminar project. The base paper provides the academic foundation upon which the proposed methodology, system design, and evaluation framework are built.

Base Paper Title: MBS Prepayment Modeling on P2P Investments: A Delinquency Risk Prediction Framework for Mortgage-Backed Securities

Authors: Naman Chachan, Dr. Prashant Lahane

Institution: MIT World Peace University, Pune, India

Format: IEEE Conference Paper Format (Double-Column, Times New Roman 10pt)

Domain: Machine Learning in Finance / Credit Risk Modeling / MBS Analytics

Year: 2025-2026

[ATTACH FIRST PAGE OF IEEE PAPER HERE]

Additional supporting reference papers utilized in this project:

- N. V. Chawla et al., SMOTE: Synthetic Minority Over-Sampling Technique, Journal of Artificial Intelligence Research, 2002.
- H. He and E. A. Garcia, Learning from Imbalanced Data, IEEE Transactions on Knowledge and Data Engineering, 2009.
- R. Emekter et al., Evaluating Credit Risk and Loan Performance in Online Peer-to-Peer Lending, Applied Economics, 2015.

Note: All referenced papers cited in this report have been published in peer-reviewed IEEE, Elsevier, ACM, or equivalent reputed journals and conference proceedings. Where applicable, papers published from 2015 onwards have been preferentially cited to ensure contemporaneous relevance.

CHAPTER 11: PLAGIARISM REPORT

11. Plagiarism Report

This section contains the plagiarism check report for the seminar report. The report was generated using a plagiarism detection tool prior to final submission.

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Declaration:

I, Naman Chachan (PRN: 1032233951), hereby declare that this seminar report titled "MBS Prepayment Modeling on P2P Investments: A Delinquency Risk Prediction Framework for Mortgage-Backed Securities" is my own original work. All sources used have been duly cited and acknowledged in the References section. The report has not been submitted for any other degree or examination.

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